

Non Linear Regression (Polynomial)

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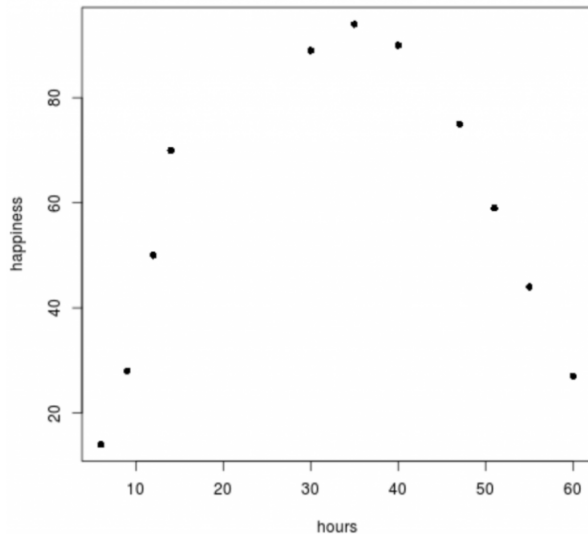
Informatics, UII

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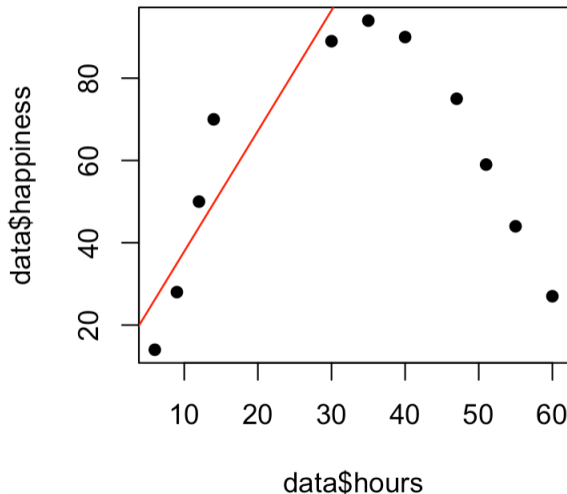
- Given the data on the number of hours worked per week and the reported happiness level (on a scale of 0-100) for 11 different people:

Hours	Happiness
6	14
9	28
12	50
14	70
30	89
35	94
40	90
47	75
51	59
55	44
60	27

Intro



Using Linear Regression?



The **straight line** does not fit the data!

We can clearly see that the data does not follow a linear pattern.

Nonlinear Regression

- Nonlinear regression is a form of regression analysis in which observational data are modeled by a function which is a nonlinear combination of the model parameters
- We will study polynomial regression, especially quadratic regression

Quadratic Regression

- We will use a quadratic function to model the data.
- The least-squares procedure can be readily extended to fit the data to a higher-order polynomial.
- In mathematical notation, a second-order polynomial or quadratic can be presented by:

$$y = \theta_0 + \theta_1 x + \theta_2 x^2$$

Quadratic Regression

The best fit function (θ_0 , θ_1 , and θ_2) can be obtained by minimizing the error (deviation).

Define the error,

$$e_i = y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2)$$

Criterion for a best fit:

$$Sr = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2)^2$$

Quadratic Regression

To find θ_0 , θ_1 , and θ_2 , set the derivative of the error (S_r) with respect to θ_0 , θ_1 and θ_2 to zero.

$$\frac{\partial S_r}{\partial \theta_0} = -2 \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2) = 0 \quad (1)$$

$$\frac{\partial S_r}{\partial \theta_1} = -2 \sum_{i=1}^n [(y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2) x_i] = 0 \quad (2)$$

$$\frac{\partial S_r}{\partial \theta_2} = -2 \sum_{i=1}^n [(y_i - \theta_0 - \theta_1 x_i - \theta_2 x_i^2) x_i^2] = 0 \quad (3)$$

Quadratic Regression

From (1),

$$\sum_{i=1}^n y_i - \sum_{i=1}^n \theta_0 - \sum_{i=1}^n \theta_1 x_i - \sum_{i=1}^n \theta_2 x_i^2 = 0 \quad (4)$$

$$n\theta_0 + \sum_{i=1}^n x_i \theta_1 + \sum_{i=1}^n x_i^2 \theta_2 = \sum_{i=1}^n y_i \quad (5)$$

Quadratic Regression

From (2),

$$\sum_{i=1}^n x_i y_i - \sum_{i=1}^n \theta_0 x_i - \sum_{i=1}^n \theta_1 x_i^2 - \sum_{i=1}^n \theta_2 x_i^3 = 0 \quad (6)$$

$$\sum_{i=1}^n x_i \theta_0 + \sum_{i=1}^n x_i^2 \theta_1 + \sum_{i=1}^n x_i^3 \theta_2 = \sum_{i=1}^n x_i y_i \quad (7)$$

Quadratic Regression

From (3),

$$\sum_{i=1}^n x_i^2 y_i - \sum_{i=1}^n \theta_0 x_i^2 - \sum_{i=1}^n \theta_1 x_i^3 - \sum_{i=1}^n \theta_2 x_i^4 = 0 \quad (8)$$

$$\sum_{i=1}^n x_i^2 \theta_0 + \sum_{i=1}^n x_i^3 \theta_1 + \sum_{i=1}^n x_i^4 \theta_2 = \sum_{i=1}^n x_i^2 y_i \quad (9)$$

Quadratic Regression

The formula (5), (7), and (9) are called **normal equations** and we can write into matrix form,

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

The above 3 equations are linear with 3 unknowns coefficients (θ_0 , θ_1 , and θ_2) which can be calculated directly from observed data, using technique such as Gauss elimination.

Polynomial Regression

The two-dimensional case can be easily extended to an m^{th} -order polynomial as

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_m x^m$$

Thus, we can easily derive the **normal equations** using the least square error method as before.

The **normal equations** can be solved using technique such as Gauss elimination or using R.

Cubic Regression

The third-order polynomial (cubic) has a form of:

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

We can derive the **normal equations** using the least square error method as before.

Cubic Regression

The **normal equations** from least square error partial derivation for cubic polynomial are:

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \sum x_i^5 \\ \sum x_i^3 & \sum x_i^4 & \sum x_i^5 & \sum x_i^6 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \\ \sum x_i^3 y_i \end{bmatrix}$$

We can solve the **normal equations** using Gaussian elimination or using R as before.

Polynomial Regression

The two-dimensional case can be easily extended to an m^{th} -order polynomial as

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_m x^m$$

Thus, we can easily derive the **normal equations** using the least square error method as before.

The **normal equations** can be solved using technique such as Gauss elimination or using R.

Polynomial Regression

The m^{th} -order polynomial has a form of:

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_m x^m$$

We can derive the **normal equations** using the least square error method as before.

Polynomial Regression

The **normal equations** from least square error partial derivation for m^{th} -order polynomial are:

$$\begin{bmatrix} n & \sum x_i & \cdots & \sum x_i^m \\ \sum x_i & \sum x_i^2 & \cdots & \sum x_i^{m+1} \\ \cdots & \cdots & \ddots & \cdots \\ \sum x_i^m & \sum x_i^{m+1} & \cdots & \sum x_i^{2m} \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_m \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \vdots \\ \sum x_i^m y_i \end{bmatrix}$$

We can solve the **normal equations** using Gaussian elimination or using R as before.

Quadratic Regression: Example

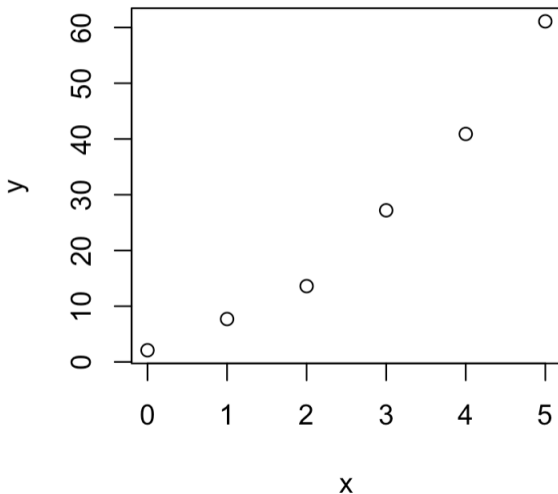
Given the data below,

x_i	y_i
0	2.1
1	7.7
2	13.6
3	27.2
4	40.9
5	61.1

Fit a second order polynomial to the data !

Quadratic Regression: Example

Data Plotting:



Quadratic Regression: Example

Solution:

x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
0	2.1	0	0	0	0	0
1	7.7	1	1	1	7.7	7.7
2	13.6	4	8	16	27.2	54.4
3	27.2	9	27	81	81.6	244.8
4	40.9	16	64	256	163.6	654.4
5	61.1	25	125	625	305.6	1527.5
Σ 15	52.6	55	225	979	585.6	2488.8

Quadratic Regression: Example

Here, we obtain:

$$m = 2$$

$$n = 6$$

$$\sum x_i = 15$$

$$\sum y_i = 152.6$$

$$\sum x_i^2 = 55$$

$$\sum x_i^3 = 225$$

$$\sum x_i^4 = 979$$

$$\sum x_i y_i = 585.6$$

$$\sum x_i^2 y_i = 2488.8$$

Quadratic Regression: Example

And we can write the normal equations:

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

into:

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{bmatrix}$$

Quadratic Regression: Example

Solving the equations:

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 152.6 \\ 585.6 \\ 2488.8 \end{bmatrix}$$

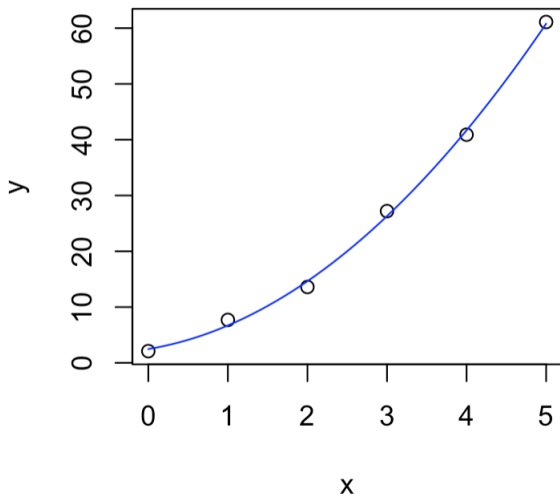
using Gauss elimination (or R), yields:

$\theta_0 = 2.47857$, $\theta_1 = 2.35929$, and $\theta_2 = 1.86071$

Therefore, the least-squares quadratic equation for this case is

$$y = 2.47857 + 2.35929x + 1.86071x^2$$

Quadratic Regression: Example



Exercise

- Given the data on the number of hours worked per week and the reported happiness level (on a scale of 0-100) for 11 different people:

Hours	Happiness
6	14
9	28
12	50
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40	90
47	75
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55	44
60	27

Fit a quadratic regression to the data above!

"Numerical analysis" by Richard L. Burden and J. Douglas Faires

"Numerical analysis" by L. Ridgway Scott

"Curve Fitting" lecture slides by Dr. Zalilah Sharer, School of Chemical and Energy Engineering, Universiti Teknologi Malaysia